

The Fidelity-Utility Paradox in Surrogate-Based RL for HVAC Control

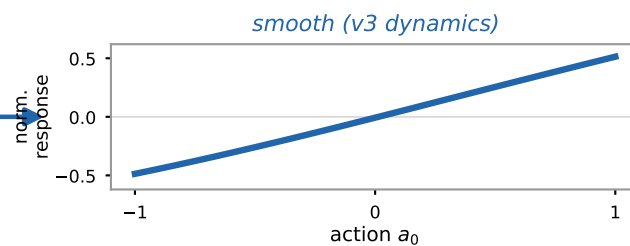
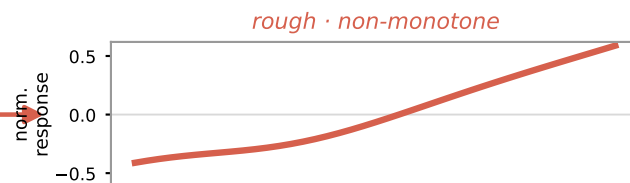
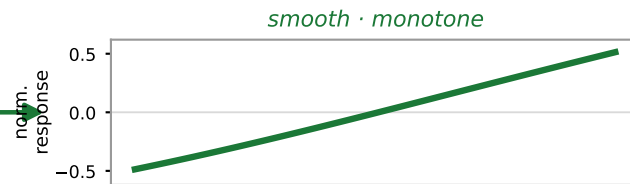
More accurate surrogate $\rightarrow$ rougher action surface $\rightarrow$ PPO collapse; hybrid separates rollout smoothness from physical censoring.

Surrogate training environment

Coarse black-box v3

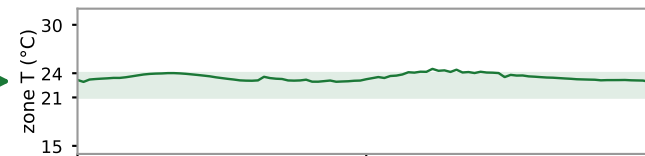
24 h RMSE 1.557 °C $\cdot m_s \approx 0.08$

Normalised action $\rightarrow$ next-temperature response

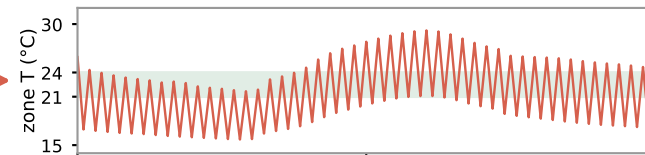


Zone temperature on the live BOPTEST runtime (24 h)

✓ **USABLE**



✗ **COLLAPSE**



✓ **ROBUST**

